



Gujarat University

M.Sc. Applied Mathematical Science

Semester - III

Name	JAYARAJ DAVID
Roll no.	12
Subject Code	AMS506PR
Subject Name	Research Methodology and Multivariate Analysis
Topic	Investigating Coping Mechanisms and Stress Factors for Students

Department of Applied Mathematical Science, Actuarial Science
and Analytics, Gujarat University

CONTENTS :

Sr.No	Topic
1	Abstract
2	Literature Review
3	Methodology • Types of Variables • Descriptive Analysis • Stress Factors
4	Part-I : Exploring Stress Factors and Stress Levels • Hypothesis (Alternate) • Results • Conclusion
5	Part-II : Investigating Coping Behaviours and Stress Levels • Introduction • Hypothesis (Alternate) • Results • Some common Coping mechanisms
6	References

ABSTRACT

- Stress is a pervasive aspect of student life, impacting both academic performance and overall well-being.
- Academic pressures, personal expectations, and societal demands converge to create a complex landscape of stressors for students.
- This study aims to uncover the intricate relationship between stress factors and coping mechanisms among students. In this research, we seek to identify the key determinants of stress and effective strategies for coping.
- Student stress factors include academic workload, peer pressure, family expectations, financial concerns, health issues, social isolation, and a lack of support. Understanding the factors that significantly affect stress levels is crucial for effective support.
- Coping mechanisms encompass various strategies employed by individuals to manage stress. The choice and frequency of coping mechanisms play a vital role in determining stress outcomes.

- This research is divided into two parts:
 - ✓ **Part I: Exploring Stress Factors and Stress Levels**
(Employing multiple linear regression analysis to explore the relationship between stress factors and stress levels. Insights are gained into the most influential stress factors and their impact on student well-being.)
 - ✓ **Part II: Investigating Coping Mechanisms and Stress Levels**
(Investigating the linear relationship between coping frequency and stress levels. This analysis sheds light on how coping behaviors contribute to mitigating student stress.)
- Thus, this research findings hope to offer insights into the most influential stress factors and the role of coping behaviors in student well-being.
- Through empirical analysis and interpretation, we aim to contribute to the discourse surrounding student well-being and offer actionable insights for the academic community.

LITERATURE REVIEW

- According to **Lazarus and Folkman (1984)**,
 - “**Psychological stress** is a particular relationship between the person and the environment that is appraised by the person as taxing or exceeding his or her resources and endangering his or her well-being” and,
 - The cognitive and behavioral strategies that an individual uses to manage a stressful situation and the concomitant negative emotions are commonly referred to as **coping strategies** (Folkman & Lazarus, 1980)
- Individuals use more or less adaptive means of coping with situations. The effectiveness of coping strategies has been shown to vary with the type of problem being faced and the degree of stressfulness of a situation (**Aldwin & Revenson, 1987**).
- According to **Feng (1992)** and **Volpe (2000)**:
 - **Stressor** is anything that challenges an individual's adaptability or stimulates an individual's body or mentality.
 - Stress can be caused by environmental factors, psychological factors, biological factors, and social factors. It can be negative or positive to an individual, depending on the strength and persistence of the stress, the individual's personality, cognitive appraisal of the stress, and social support.

- In *Akgün, S., & Ciarrochi, J. (2003)*,
 - it was studied how **learned resourcefulness moderates the relationship between academic stress and academic performance**. It suggests that the negative impact of stress was moderated by learned resourcefulness.
 - Specifically, academic stress had a significant negative effect on the academic performance of low resourceful students, but had **no effect on high resourceful students**.
- A survey was conducted to investigate the sources of stress among college students in Taiwan. [*Kai-Wen & et al (2009)*].
 - In this study it was revealed that the most significant stressors in students were '**Family Expectations**' & '**Financial situation**'
- In *Ji, H., & Zhang, L. (2011)*,
 - Another such survey conducted on 734 college students indicates that the **mental stresses** of college students have **positive correlations** with study conditions, employment situations, economic conditions, and personal factors.

METHODOLOGY

TYPES OF VARIABLES :

- The questionnaire employs a **Likert scale of 5 (1 to 5)**.
- The variables and their types are as follows:
 1. **Age** – Quantitative variable (can be used for categorization)
 2. **Gender** – Nominal Variable (can be used for categorization)
 3. **Education** – Nominal Variable (can be used for categorization)
 4. **Overall Stress** – Ordinal variable (Dependent variable)
 5. **Academic Workload** – Ordinal variable (Independent variable) i.e X_1
 6. **Peer Pressure** – Ordinal variable (Independent variable) i.e. X_2
 7. **Social isolation** – Ordinal variable (Independent variable) i.e. X_3
 8. **Financial Issues** – Ordinal variable (Independent variable) i.e. X_4
 9. **Health Issues** – Ordinal variable (Independent variable) i.e. X_5
 10. **Family expectation** – Ordinal variable (Independent variable) i.e X_6
 11. **Lack of support** – Ordinal variable (Independent variable) i.e X_7
 12. **Coping Frequency** – Ordinal variable (Independent variable)

BASIC ANALYSIS

- *“Descriptive analysis is a statistical method used to summarize and present data in a clear, understandable way. It involves organizing, summarizing, and visualizing data to provide insights into its key characteristics.”*
- It **does not involve making inferences** or drawing conclusions about a larger population; instead, it focuses on describing the data you have in a **concise and meaningful manner**.
- Using the categorical variables: **Age, Gender** and **Education level** to analysis how the sample is divided as follows:

<u>Gender</u>		
Choice	Frequency	Percent
Female	54	49.5%
Male	55	50.5%
Total	109	100.0

- From the given table, it can be observed that the sample is divided into **Male** and **Female** as **49.5%** and **50.5%**.

<u>Age</u>		
Choice	Frequency	Percent
<18	7	6.4
18-21	64	58.7
22-25	38	34.9
Total	109	100.0

- Now, in the frequency distribution for **Age**, it can be observed that the sample is divided into the age groups of (**<18**), (**18 – 21**), and (**22 – 25**) as **6.4%**, **58.7%** and **34.9%** respectively.

Education Level		
Choice	Frequency	Percent
High School/ Diploma	8	7.3
Undergraduate level	71	65.1
Postgraduate level	30	27.5
Total	109	100.0

- Now, in the frequency distribution for **Education Level**, it can be observed that the sample obtained is divided into the groups of **High School / Diploma**, **Undergraduate**, and **Postgraduate** as **7.3%**, **65.1%** and **27.5%** respectively.

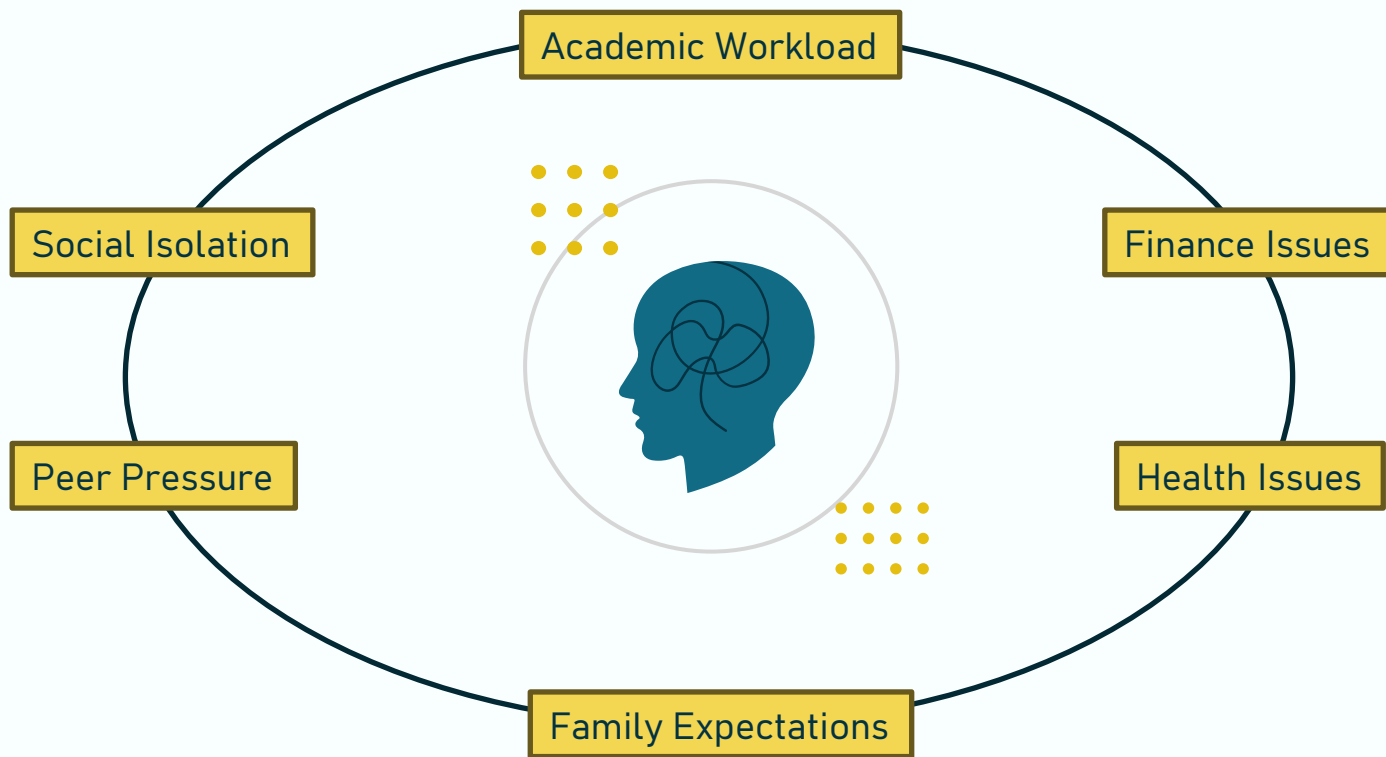
Part - I

Exploring Stress Factors and Stress Levels

- In this section, we will start peeling back the layers of student stress, uncovering the key factors that significantly influence their stress levels.
- To execute this, we will employ a **regression model** to find out which stressors have a significant impact on the overall stress level in students.
- By identifying the stress, we offer valuable insights that can inform the **development of targeted interventions** by educational institutions and support services.
- Before we show the results, let's understand what multiple linear regression is!



STRESS FACTORS UTILIZED IN THIS STUDY



HYPOTHESIS :

- H_{1a} : There is a significant impact of stress caused due to **academic workload** on overall stress.
- H_{1b} : There is a significant impact of stress caused due to **peer pressure** on overall stress.
- H_{1c} : There is a significant impact of stress caused due to **Social isolation** on overall stress.
- H_{1d} : There is a significant impact of stress caused due to **Financial Issues** on overall stress.
- H_{1e} : There is a significant impact of stress caused due to **Health Issues** on overall stress.
- H_{1f} : There is a significant impact of stress caused due to **Family Expectations** on overall stress.
- H_{1g} : There is a significant impact of stress caused due to **Lack of support** on overall stress.
- H_2 : The regression model constructed is significant.

RESULTS :

- Using the data we obtained with our questionnaire, we can simply input the data into SPSS software and get the following outputs:

Model Summary				
Model	R	R Square	Adjusted R Square	Std. Error of the Estimate
1	0.799	0.639	0.614	0.8073

- Here, since the value of R-Square is 0.639, we can say that the multiple linear regression model constructed is able to explain about 63.9% of the variance.

- The ANOVA table obtained is as follows:

<u>ANOVA</u>						
Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	116.492	7	16.642	25.537	.000
	Residual	65.819	101	.652		
	Total	182.312	108			

- Here, since the significance level is less than **0.05**, we can conclude that this model is **significant**.

Model		Unstandardized Coefficients		Standardized Coefficients	Sig.
		B	Std. Error	Beta	
1	(Constant)	-.236	.279		.400
	acad_wl	.352	.081	.323	.000
	peer_pressure	.172	.082	.160	.039
	Social_isol	.032	.079	.033	.683
	finance	.328	.080	.345	.000
	health	.081	.066	.087	.223
	fam_expect	.171	.076	.167	.027
	lack_sup	.006	.067	.006	.931
a. Dependent Variable: ovr_stress					

Let's look at the hypothesis!

Hypothesis	Significance	Significant or not?
H _{1a} : There is a significant impact of stress caused due to academic workload on overall stress.	0.000	Significant
H _{1b} : There is a significant impact of stress caused due to peer pressure on overall stress	.039	Significant
H _{1c} : There is a significant impact of stress caused due to Social isolation on overall stress.	.683	Not Significant
H _{1d} : There is a significant impact of stress caused due to Financial Issues on overall stress.	.000	Significant

Hypothesis	Significance	Significant or not?
H_{1e} : There is a significant impact of stress caused due to Health Issues on overall stress.	.223	Not Significant
H_{1f} : There is a significant impact of stress caused due to Family Expectations on overall stress.	.027	Significant
H_{1g} : There is a significant impact of stress caused due to Lack of support on overall stress.	.931	Not Significant
H_2 : The multiple linear regression model constructed is significant .	.000	Significant

CONCLUSION

- From the results obtained, we conclude that from the seven factors that we included in this study: **“Academic Workload”**, **“Peer Pressure”**, **“Financial Concerns/Issues”**, **“Family Expectations”**; are the factors that have a significant impact on the Stress.
- By observing the Standardized coefficients of the variables, we can see that out of the four factors, **“Academic workload”** and **“Financial Concerns”** have a significantly positive impact on the **Overall Stress of a student** i.e.
 - It suggests that as the academic workload increases, students are more likely to experience higher stress levels.
 - Students facing financial worries are more likely to experience elevated stress.
- These two factors are followed by: **“Peer Pressure”** and **“Family Expectations”** which, although have a lesser impact than the above two factors, have a significant positive impact on the Overall Stress as well!



CONCLUSION

- However, social isolation, health concerns, and the perceived lack of support do not appear to be statistically significant contributors to stress levels in our study.
- The multiple regression equation obtained is as follows:

$$Ovr_Stress = -0.236 + 0.352 * X_1 + 0.172 * X_2 + 0.032 * X_3 + 0.328 * X_4 + 0.081 * X_5 + 0.171 * X_6 + 0.006 * X_6$$

- This equation can be further used for predicting purposes as well.
- These results underscore the importance of addressing these factors to reduce stress and enhance the well-being of students.
- The findings obtained in this study can be then used by academic/ educational organizations to develop support systems for the students, which might greatly help in maintaining the mental health of students.

Part - II

Investigating Coping Mechanisms and Stress Levels



- In this second part of our research, we turn our focus to the **coping mechanisms employed by students** in response to the stresses of academic life.
- Coping mechanisms represent a diverse set of strategies and behaviors individuals use to **manage, alleviate, or adapt** to stressful situations.
- The frequency and effectiveness of these coping strategies play a crucial role in shaping students' overall stress experiences.
- To execute this, we will employ a **simple linear regression model** to find out if utilizing coping mechanism actually has a significant impact on the overall stress level in students.

- By exploring coping behaviors, we aim to uncover valuable insights that can inform the development of effective interventions and resources to support student well-being.
 - In this part of study, we hope to answer the question: **“How do these coping behaviors relate to their levels of stress?”**
-

HYPOTHESIS :

- H_1 : There is a significant impact of coping frequency (The number of times, a student uses some technique to manage stress) on the overall stress of a student.

RESULTS :

- Using the data we obtained with our questionnaire, we can simply input the data into SPSS software and get the following outputs:

Model Summary				
Model	R	R Square	Adjusted R Square	Std. Error of the Estimate
1	0.616	0.379	0.373	1.0287

- Here, since the value of R-Square is 0.379, we can say that the linear regression model constructed is only able to explain about 37.9% of the variance.

- The ANOVA table obtained is as follows:

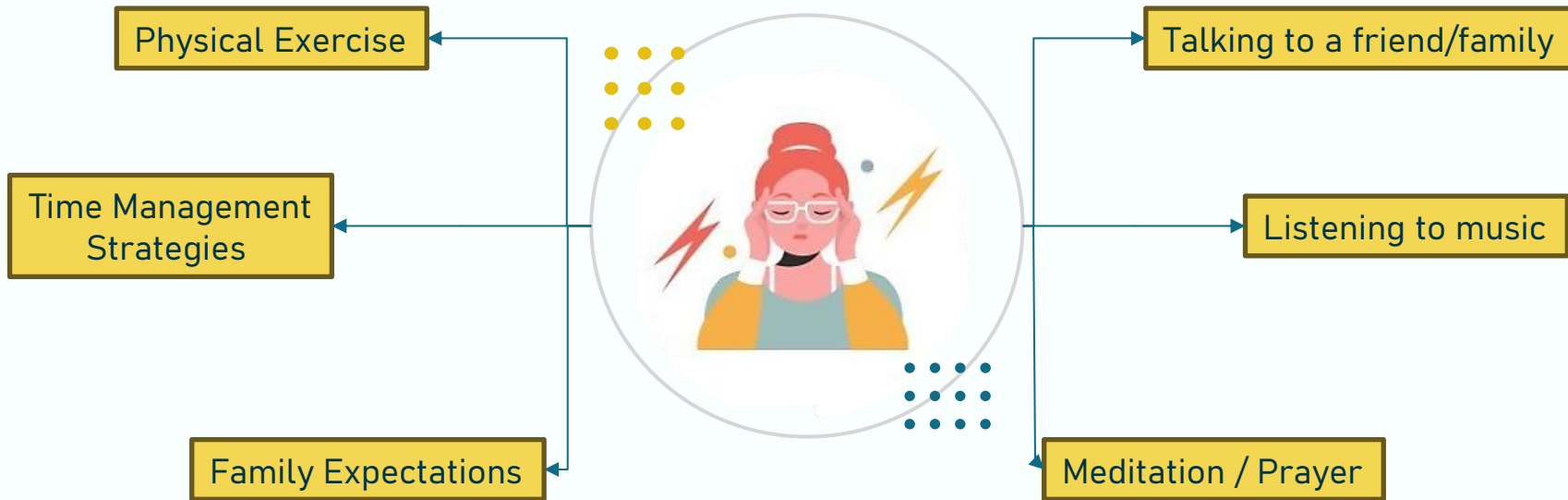
<u>ANOVA^a</u>						
Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	69.090	1	69.090	65.293	.000 ^b
	Residual	113.222	107	1.058		
	Total	182.312	108			

- Here, since the significance level is less than **0.05**, we can conclude that this model is **significant**.

Model		Unstandardized Coefficients		Standardized Coefficients	Sig.
		B	Std. Error	Beta	
1	(Constant)	5.247	.266		.000
	Coping_freq	-.795	.098	-.616	.000
a. Dependent Variable: ovr_stress					

- Here, from the coefficients table, it is clear that Coping frequency has a significant impact on Overall stress of students.
- Therefore, our null hypothesis H_1 is rejected.
- Also, notice from the standardized coefficient beta ($=-0.616$), the coping frequency has significantly negative impact on overall stress i.e. increase in coping frequency causes the overall stress to reduce for our sample of students.

SOME COPING MECHANISMS !



References:

- [1] Akgün, S., & Ciarrochi, J. (2003). Learned Resourcefulness Moderates the Relationship Between Academic Stress and Academic Performance. *Educational Psychology*, 23, 287-294. <https://doi.org/10.1080/0144341032000060129>
- [2] Aldwin, C., & Greenberger, E. (1987). Cultural differences in the predictors of depression. *American Journal of Community Psychology*, 15(6), 789–813. <https://doi.org/10.1007/bf00919803>
- [3] Banks, J., & Smyth, E. (2015). 'Your whole life depends on it': Academic stress and high-stakes testing in Ireland. *Journal of Youth Studies*, 18(5), 598-616. <https://doi.org/10.1080/13676261.2014.992317>
- [4] Ji, H., & Zhang, L. (2011). Research on college students' stresses and coping strategies. *Asian Social Science*, 7(10), 30. <https://www.ccsenet.org/journal/index.php/ass/article/view/12451>
- [5] Kai-Wen, Cheng. "A study of stress sources among college students in Taiwan." *Journal of Academic and Business Ethics* 2 (2009): 1. <https://citeseerx.ist.psu.edu/document?repid=rep1&type=pdf&doi=725729d04cde923ea09dca38d8702c52250a7342>



References:

- [6] Robotham, D. (2008). Stress among higher education students: Towards a research agenda. *Higher Education*, 56(6), 735-746. <https://doi.org/10.1007/s10734-008-9137-1>
- [7] Sarros, J. C., & Densten, I. L. (1989). Undergraduate student stress and coping strategies. *Higher Education Research & Development*, 8(1), 47-57. <https://doi.org/10.1080/0729436890080105>

**THANK
YOU!**